

INTRODUCTION

The purpose of this manual is to supply essential information to customers, contractors, builders, architects, engineers, and others concerned with the planning of electrical installations.

All information given in this manual is to be used in conjunction with the Lake Mills Light and Water’s Rate Tariffs and Electric Rules on file with the Public Service Commissions of Wisconsin. In issuing this manual, the Lake Mills Light and Water is in no way relieving the customer or contractor of his or her responsibility to install the wiring in accordance with the applicable local and Wisconsin State Electrical Codes, or to maintain the wiring and equipment in a safe operating condition. Electrical wiring should only be done by those who have been trained on safety concerns, wiring techniques and code requirements. Lake Mills Light and Water cannot accept any responsibility for the condition of the customer's wiring and equipment. The information contained herein is general and may not cover all conditions. In all cases, safety will be paramount in the decision to make any exceptions to our procedures. For new or special cases not specifically covered in this manual, Lake Mills Light and Water should be consulted.

This edition of the **Electrical Service Manual** supersedes any and all previous editions.

LML&W reserves the right to refuse service if a customer does not comply with these rules or if a dangerous electrical condition exists on a customer’s premises.

LML&W requests that contractors, developers and homeowners contact LML&W early in the design process to ensure all issues are resolved prior to construction.

Before any new electric service is energized the electrical contractor, building owner, or general contractor must be on site to take responsibility for the newly energized service or LML&W crews must have safe access to the service’s main disconnect.

Key Contacts:

- Paul Hermanson**, Director of Public Works..... 920-648-4026
phermanson@ci.lake-mills.wi.us
- Chris Evenson**, Lead Foreman..... 920-648-4026
- Josh Hergert**, Lead Foreman..... 920-648-4026

TO OBTAIN ELECTRIC SERVICE

LML&W requires an application for electric service be completed and returned to the appropriate city or town Building Inspector prior to connecting a service. Each application shall include an itemized copy of the connected load, including lights, motors and appliances. To upgrade an existing service or for additional service facilities, customers must contact the Building Inspector to determine if permits or inspections are needed. A copy of the application is available at the Lake Mills municipal building 200 Lake Street, Lake Mills, WI.

The applicant agrees to pay any required contribution in aid of construction. The utility may require that the contribution in aid of construction be paid in advance of construction or may, at the utility's option, offer customers an installment payment plan. For more detail or information, refer to the latest LML&W Rate Tariffs and Electric Rules, filed with the Public Service Commission of Wisconsin (PSC).

The utility shall provide one standard overhead service drop or one standard underground service lateral per premise at no charge to a customer. For more details, refer to the appropriate service entrance diagram in this service manual and the latest LML&W Rate Tariffs and Electric Rules as filed with the Public Service Commission of Wisconsin (PSC). A "frost adder" will be in effect when frost reaches a depth of 6" as well as a "rock trenching adder" for all underground service laterals.

TEMPORARY SERVICE FOR CONSTRUCTION

LML&W will install temporary services to provide electricity during construction. It is preferred that temporary services remain in place for no more 6 months unless prior approval is gained. **Please contact LML&W at 920-648-4026 prior to installing a temporary electric service.**

Temporary Service requirements for construction are:

1. The customer will agree to reimburse the utility for its expenditures in extending electric service.
2. Temporary service shall be given to a customer connection only when constructed in accordance with this service manual. The post supporting the unit shall be located as near as possible to the location of permanent service to the building. Abnormal conditions involving compliance with the foregoing provision will be cleared with the utility and permission granted by the utility prior to locating the customer connection.
3. All temporary service shall be maintained in a safe manner in order to keep the utility harmless from injury to persons or property. The service shall remain temporary only for a reasonable time and must be made permanent when the utility directs such action.
4. Should the customer elect to receive permanent service, the installation charges for extension of new electric service will apply. Credit shall be given for the payment already made for that portion of the temporary service facilities that can be used for permanent service without modification.
5. Design and layout of temporary service must be approved by LML&W.

The utility will also extend temporary electric service to fairs, carnivals and like short-time gatherings and uses (not including short-time uses in the nature of auxiliary, stand-by or seasonal use) under the following rules:

1. The customer will agree to reimburse the utility for its expenditures in extending service.
2. The cost of extending service shall include all items of labor and materials, with the customary overhead charges, necessary to furnish the customer with the service requested. It shall also include any costs involved in the dismantling of materials and their return to stock. Where materials dismantled have a salvage value, the cost of extending service will be credited with such salvage value.
3. All energy will be measured at one standard voltage at some convenient point designated by the utility.
4. The customer will make the necessary arrangements and provide for the necessary equipment in the event more than one voltage is required.
5. The cost of all construction (labor and materials) necessary to distribute energy on the premises occupied by the customer will be borne by the customer.
6. The utility may require the customer to make an advance deposit sufficient to cover the costs of extending service and the estimated bill for energy.
7. The rates applicable in the area where temporary service is rendered shall be applied in determining the customer's bill.

LOCATING A NEW ELECTRIC OVERHEAD & UNDERGROUND SERVICE

LML&W shall, in all cases, designate the location where the LML&W service will terminate. LML&W uses the following guidelines when sighting a new service entrance location:

- The preferred residential and small commercial service spot is within six feet of the corner of the living space nearest the LML&W source.
- The location shall minimize LML&W investment in the service and meter making.
- LML&W facilities shall not be located in areas that are likely to cause conflicts with the following:
 - Enclosed areas, patios, porches, doorways, etc.
 - Concrete areas (patios and driveways)
 - Carports, fenced-in areas, pools and hot tubs, wells
 - Clearance problems/codes, etc.
 - Common walls with gas and electric meters, LP tanks and lines and above/below ground fuel tanks
 - Potentially hazardous, high traffic areas (sidewalks, driveways, alleys)
 - Installation and future maintenance costs
 - Remote locations that would make meter reading or maintenance difficult (fenced yards)
 - Septic systems and their alternate fields
 - Future buildings or additions and Grade changes
 - Buried customer facilities
- An overhead service is limited to a maximum of 150' because of voltage drop, clearance and conductor tension constraints.
- If the customer prefers to locate the service entrance somewhere other than the LML&W designated location, the customer has that option, but may be subject to a special facilities payment for the added materials and labor.
- Variations – unusual house designs and variations in LML&W system construction will be encountered in the field. Each situation should be addressed individually, based on the guidelines included in this section.

ELECTRIC SERVICE CHECKLIST

Prior to LML&W scheduling a service connection, the customer is responsible to confirm that:

- ✓ A written application for electric service has been submitted to the utility.
- ✓ All fees or charges to install a service are paid in full,
- ✓ All necessary easements are signed,
- ✓ Service lateral route is clear of dirt piles, trees, buildings, other obstructions,
- ✓ Service entrance wiring is completed,
- ✓ Electrical inspection is complete, and
- ✓ Grade is within four inches of final grade.

VOLTAGES

Lake Mills Light and Water has dual primary distribution voltages of 4,160 VAC and 24,940 VAC.

LML&W standard secondary voltages are:

- 120/240 VAC 3-wire single phase
- 120/208 VAC 4-wire wye three phase
- 277/480 VAC 4-wire wye three phase

Secondary voltages no longer provided as new services:

- 240 Volt, 3Ø, 3-Wire Service, Grounded B Phase
- 480 Volt, 3Ø, 3-Wire Service

For some classes of service, especially three phase, availability is also dependent upon the customer's connected load. LML&W must be contacted for cost and availability of service.

For residential services rated 320 amps or less, service voltage shall be 120/240 VAC 3-wire single phase. Residential services rated greater than 320 amps, service voltage can be either 120/208 VAC 4-wire wye three phase or 277/480 VAC 4-wire wye three phase, dependent upon the customer's connected load.

For commercial services rated 200 amps or less, service voltage can be either 120/240 VAC 3-wire single phase, 120/208 VAC 4-wire wye three phase, or 277/480 VAC 4-wire wye three phase, dependent upon the customer's connected load. For commercial services rated greater than 200 amps, service voltage can be either 120/208 VAC 4-wire wye three phase or 277/480 VAC 4-wire wye three phase, dependent upon the customer's connected load.

METERING

Lake Mills Light and Water will supply all electric meters. LML&W's general standard requirement for a residential single phase service rated at 320 amps continuous is a plug-in meter socket. For approved meter socket please see the service entrance diagrams included in this service manual.

For residential services rated 320 amps or less, LML&W does not require current or potential transformers. For commercial services rated 400 amps or greater, current and potential transformers may be required. Contact LML&W for more details. LML&W will furnish the current and potential transformers and complete all the necessary meter wiring. The customer must furnish an approved metering cabinet, meter socket, current and potential mounting provisions, and test switches.

Metering equipment may not be located on LML&W's facilities. All meters must be placed on the exterior of any dwelling or facility, must be protected from potential damage, and must be easily accessible for reading and maintenance.

TRANSFORMERS

Lake Mills Light and water will supply all transformers. Transformers may be overhead or padmount for use with underground service. LML&W will work with customers to ensure that location of transformers meets or exceeds all necessary clearances as outline in state and national safety codes. Transformers must be accessible by truck, when possible, for maintenance and replacement reasons.

The utility shall provide standard design transformers necessary to serve the customer's load at no charge.

A standard design transformer is a transformer with capacity less than or equal to 300 kVA. If a customer requests or requires additional capacity, the utility shall add to the total cost of installation a charge equal to the cost of the necessary transformer(s) less a credit for the cost of the maximum capacity standard transformer.

Customers will be charged for the pad or basement for underground transformers.

SERVICE ENTRANCE DIAGRAMS

This series of Service Entrance Diagrams is meant to show typical or common service entrances and metering set-ups. Variations to these outlines may be necessary in some applications, but all service entrances must conform to applicable codes, pass electrical inspection and must be approved by LML&W.

- Unless otherwise indicated, all items shown on the diagrams are to be furnished and installed by the customer.
- Meters and metering transformers will be furnished, installed and maintained by LML&W.
- Only meter mounting devices and service termination equipment approved by LML&W are to be used.
- The listing of equipment in this manual is not an endorsement or indication of suitability, but only that it is acceptable for installation of our metering equipment and termination of our service lateral conductors as required. It is the responsibility of the customer or their electrical contractor to verify that the equipment is suitable for the installation and that it is installed in accordance with all applicable codes.
- Electrical wiring should only be done by those who have been trained on safety concerns, wiring techniques, and code requirements. LML&W cannot inspect your wiring or provide electrical code interpretations. This is to be done by State-Certified Electrical Inspectors.
- Electrical contractors are expected to acquaint themselves with the plans of other trades on the premises being wired so that the meter can be located in accordance with LML&W Electric Service Rules and Regulations.
- Transfer equipment used with stand-by power plants or generators shall be suitable for intended use and be designed and installed as to prevent the inadvertent interconnection of normal and stand-by energy sources. Transfer equipment shall be installed such that it is located electrically on the load side of the meter. All transfer schemes shall be submitted to the LML&W office for acceptance before installation.
- The use of approved material, tested and listed by a nationally recognized testing laboratory, and approved methods of installation are requirements of the Wisconsin State Electrical Code by the Wisconsin State Electrical Code Section of the Department of Commerce, and by OSHA for employers, for customer owned electrical service equipment.
- This manual only covers secondary services. For primary services, contact LML&W.

GENERAL REQUIREMENTS FOR UNDERGROUND SERVICE

Space Requirements

1. The unobstructed space required in front of termination compartments, transockets and metering equipment shall be as defined by the "Working Space About Electrical Equipment", Section 110.26, of the National Electrical Code. This unobstructed space shall extend from the floor or ground to a minimum height of 6'-6". For equipment higher than 6'-6" this space shall extend to the top of the equipment.

Service Laterals and Entrances

1. Service laterals are furnished and installed by LML&W.
2. Service laterals will generally be terminated in a meter pedestal, meter socket, transocket or termination compartment. Customers shall obtain information from LML&W relative to the proper location of the service lateral and termination equipment. Such information will be furnished in writing upon request.
3. The preferred termination of service laterals is on the outside of a building (or structure). Service laterals that terminate inside of a building shall terminate immediately upon entering as shown in this manual. The raceway for service laterals under or thru a building shall not extend longer than 8' after penetrating the outside wall.
4. For services, generally 2000 Amperes and larger, and with LML&W approval, the service lateral may be terminated in the customer's switchgear or on the customer's bus duct. This type of installation is further restricted by the following:
 - a. Customer shall provide all necessary plastic conduit encased in 3" of concrete for the extension of the cable within his building (see Note 3 above).
 - b. Customer shall obtain LML&W acceptance of the switchgear terminal and C.T. compartment (if any) design. Three copies of the switchgear plans shall be submitted to LML&W for acceptance. The plans must show the general layout of the switchgear, the switchgear location within the building, and the specific details of the termination compartment and any C.T. compartment(s). Specific details such as compartment size, bus spacing and drilling dimensions, other clearances, bus & bus support configuration, barrier location, sealing provisions, and materials used are to be included.
5. Only meter mounting devices and termination equipment rated to accept underground conductor, approved and listed in this manual are to be used. All equipment must be listed by a nationally recognized testing laboratory, e.g. UL, installed and used as per the listing and the manufacturer's instructions.
6. Where metallic rigid conduit or IMC is used for the service lateral raceway the ends are to be threaded and threaded couplings, connectors or locknuts are to be used. Slip type fittings shall not be used.
7. For outdoor service terminations with direct buried lateral, the horizontal portion of the conduit shall extend through a minimum of 12" of undisturbed earth where practicable, defined as follows:
 - a. Undisturbed earth: The area not excavated for the construction of the building. (May be compacted fill.)
 - b. Fill area: The area between the building wall and the undisturbed earth. (May vary from a few inches to several feet or more).
8. Services to individual dwelling units shall consist of one service lateral and one meter.

Grounding

1. Grounding shall be in accordance with all applicable codes.
2. Stainless steel (A.I.S.I. 302, 304 or 416) ground rods are recommended where ground rods must be installed in the vicinity of buried galvanized rigid conduit to minimize the possibility of corrosion.
3. The grounding electrode conductor shall not be run through the meter pedestal, meter socket, transocket, or termination compartment, unless a separate raceway is provided. The grounding electrode conductor may terminate on the neutral bus of: the termination compartment for multi-metered installations provided it does not interfere with the installation of the service lateral conductors, or a meter pedestal equipped with a main disconnect and listed as Service Equipment.
4. Ground rods and grounding electrode conductors shall not be located in front of meter pedestals or wiring troughs.
5. Bonding to other systems shall not be done on or within a metering enclosure unless a means of bonding, intended for inter-system bonding, is furnished as part of the metering enclosure (PSC 114.099(C)).

GENERAL REQUIREMENTS FOR OVERHEAD SERVICE

Service Drops, Attachments, Supports, and Entrances

1. All service drops are furnished and installed by LML&W.
2. In all installations except those stated in the General Service and Extension Rules, on file with the PSC, the customer shall verify the proper location of the service head and service drop from LML&W, which will furnish this information in writing upon request.
3. In all cases, an adequate support for the service drop attachment and service drop shall be provided by the customer. LML&W furnishes attachments and will provide information as to the pounds of pull that the service drop support must be designed for or withstand.
4. Typical installations for various service voltages and ampere capacities are shown in succeeding pages. Unusual installations require special consideration and LML&W shall be consulted in all cases.
5. Only meter mounting devices and termination equipment rated to accept overhead conductors, approved and listed in this manual are to be used. Equipment must be listed by a nationally recognized testing laboratory, e.g. UL, installed and used per the listing and the manufacturer's instructions.
6. The service head is to be located at a height such that the service drop conductors can be dead-ended below the service head and still maintain proper clearance above ground, but they are not to be installed higher than 30' above ground.
7. When a service mast is used to support the service drop, only the service drop is to be attached to it. Other utilities such as telephone or cable TV shall not be attached to the service mast.

Grounding

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2. Stainless steel (A.I.S.I. 302, 304, or 416) ground rods are recommended where ground rods must be installed in the vicinity of buried galvanized rigid conduit, to minimize the possibility of corrosion.
3. The grounding electrode conductor shall not be run through the meter socket or transocket.
4. Bonding to other systems shall not be done on or within a metering enclosure unless a means of bonding, intended for inter-system bonding, is furnished as part of the metering enclosure (PSC 114.099(C)).

TEMPORARY SERVICES - UNDERGROUND AND OVERHEAD

NOTES:

1. All materials shall be furnished and installed by the owner unless noted otherwise. The utility shall specify the location of temporary service poles. Care must be taken to avoid the permanent service route. Potential clearance problems must also be avoided.
2. The meter sockets shall be 100 or 200 Amp rated. They shall be a ringless design with bypass horns and UL approved. LML&W will provide and install the meter.
3. The switchgear shall be installed per code. This requires the switchgear to be weatherproof or protected from the elements. It also requires GFI protection on all outlets unless OSHA-required electrical inspections are done regularly (Difficult to comply with).
4. The temporary service pole shall be at least 4-inches in diameter, provided and installed by customer.
5. Ground as required by code. Two ground rods are required. Also, a bond to a metallic water pipe may be required, if available. The ground conductor shall be terminated in the switchgear and not in the meter sockets,
6. Brace and stakes made out of 2 x 4's.
7. For residential 100 Amp temporaries, the conductor must be at least #4 copper or #2 aluminum. For residential 200 Amp temporaries, the conductor must be at least 2/0 copper or 4/0 aluminum. Service entrance conductors shall extend at least 30" beyond the service head to permit proper connection to LML&W conductors.
8. Temporary services are intended only for a short use. If the temporary service is expected to be used longer than one year, see Sections 2 or 3 for permanent installation instructions.
9. LML&W will provide service conductors and make all connections to the LML&W system.
10. Temporaries must be high enough to maintain basic code ground clearances. Refer to basic clearances-services in this manual for more details.
11. Used equipment, such as 60 Amp meter sockets and old indoor 60 Amp fuse panels, are not acceptable for temporaries. If LML&W must make return trips because of clearance problems or unsafe equipment, there may be additional charges.
12. Option #5 shows how to install a permanent meter pedestal and breaker panel and how to avoid using a temporary service. The breaker panel must either be weatherproof or protected from a wet or damp environment. The basement walls must be backfilled and adequately tamped for LML&W to run the permanent underground service. Another option is to use approved flexible conduit between the meter pedestal and the breaker box. See the appropriate service sketch for details on permanent underground services.